

STATISTICAL ANALYSIS ON NETFLIX VIEWERSHIP

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Introduction

Netflix is the world's leading entertainment service with over 260 million total paid subscribers reaching over 190 countries. It provides access to top-in-class TV series, movies, documentaries, animated shows, feature films and even games. It is streaming in over 30 languages and is available in 190 countries. Its dominance is facing uncertainty, with strong competitors like Disney+, Hulu, and Amazon Prime in close pursuit. Notably, Netflix charges the highest prices compared to its rivals. (Duarte, 2024). While this is not an immediate issue, a future challenge may arise if current subscribers feel uninspired by Netflix's content offerings.

Objectives

The primary objective of this project is to conduct a thorough data analysis on Netflix, employing data from its vast repository of movies and TV series to enhance business strategies, particularly by improving Netflix's Recommendation Systems. This will involve analyzing viewership statistics provided by Netflix, spanning from January to June 2023, in tandem with IMDb ratings.

Research questions

1. What factors influence the viewership of a movie or TV show on Netflix?
2. What genre garners the highest preference among viewers?
3. Which combination of content type and genre leads to high viewership?
4. How can the insights gained from this analysis be utilized to develop business strategies tailored to viewer preferences?

Research hypotheses

In our research, we hypothesize that viewership on Netflix is significantly influenced by a combination of content type, genre, duration, number of awards won and nominated, IMDb ratings and global availability.

Data sources

The data sources utilized in this project included Netflix, IMDb and Wikipedia.

Variable Terminology

Variable Name	Description	Type	Explanatory/Dependent
Title	Title of the movie/TV show	Qualitative	Explanatory
Type	Type of the streaming content (Movie/TV show)	Qualitative	Explanatory
Genre	Genre of the movie/TV show	Qualitative	Explanatory
Release Year	Original release year	Qualitative	Explanatory
Duration	Runtime of the streaming content in minutes	Quantitative	Explanatory
No. of Seasons	How many seasons of the TV show have been released?	Quantitative	Explanatory
No. of Episodes	Total number of episodes of the TV show	Quantitative	Explanatory
Awards Won	Number of awards won	Quantitative	Explanatory
Award Nominations	Number of award nominations	Quantitative	Explanatory
IMDb Rating	Audience ratings collected from IMDb	Quantitative	Explanatory
Global Availability	Is the media content accessible globally	Qualitative	Explanatory
Netflix Original	Is the streaming content produced by Netflix	Qualitative	Explanatory
Original Language	Original language in which the movie/TV show was released	Qualitative	Explanatory
Viewership	Total number of hours the content has been viewed on Netflix	Quantitative	Response

Limitations

The viewership data collected from Netflix's engagement report unfortunately cannot provide a complete picture since the viewership data only accounts for views during the period of January to June of 2023. There may be some skewness towards Netflix titles that were released between the same period. (Netflix, 2023)

Methodology

The following statistical methods used in our research are as follows:

1. Multiple Linear Regression
2. One way ANOVA
3. Two-way ANOVA
4. Chi-squared test for Goodness of Fit

Test 1: Multiple Linear Regression

In our analysis, we utilized viewership (measured in hours) as our dependent variable, while considering content type, genre, and global availability as categorical predictors. Additionally, duration, number of awards won, number of awards nominated, and IMDb rating were included as continuous predictors. To evaluate multicollinearity among predictors, we examined the correlation matrix. We also conducted the test for Normality by plotting the P-P plot. For assessing the homoscedasticity of residuals and identifying outliers, we created a scatter plot. The coefficient of determination (R-squared) was calculated to determine the proportion of variance in viewership explained by the predictors in the regression model.

The results of the Multiple Linear Regression (MLR) test are presented below:

Table 1: Dummy Variables

Variable Creation	
	Label
Type_1	Type=Movie
Type_2	Type=TV Series
Genre_3	Genre=Action
Genre_4	Genre=Anime
Genre_5	Genre=Comedy
Genre_6	Genre=Drama
Genre_7	Genre=Horror
Genre_8	Genre=Thriller
Global_Availability_9	GlobalAvailability=No
Global_Availability_10	GlobalAvailability=Yes

Source: IBM SPSS Statistics 29.0

In our analysis, we incorporated dummy variables to represent the categorical predictors content type, genre, and global availability. The reference variables were set as follows: "Movie" for the category Type, "Action" for Genre, and "Yes" for Global availability, respectively.

Table 2: Model Summary of Multiple Linear Regression

Model Summary ^b					
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.582 ^a	.339	.307	13764952.694	1.014

a. Predictors: (Constant), GlobalAvailability=No, Genre=Horror, Award Nominations, Genre=Thriller, Type=TV Series, Genre=Comedy, Genre=Drama, IMDb Rating, Genre=Anime, Awards Won, Duration

b. Dependent Variable: Viewership

Source: IBM SPSS Statistics 29.0

Model summary shows the correlation between the variables, the coefficient of determination (R^2) and the Durbin-Watson statistics.

Netflix Viewership Analysis

Table 3: ANOVA of Multiple Linear Regression

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	2.194E+16	11	1.995E+15	10.528	<.001 ^b
	Residual	4.282E+16	226	1.895E+14		
	Total	6.476E+16	237			

a. Dependent Variable: Viewership

b. Predictors: (Constant), GlobalAvailability=No, Genre=Horror, Award Nominations, Genre=Thriller, Type=TV Series, Genre=Comedy, Genre=Drama, IMDb Rating, Genre=Anime, Awards Won, Duration

Source: IBM SPSS Statistics 29.0

The ANOVA table output is utilized to assess the significance of the overall regression model.

Table 4: Coefficients of Multiple Linear Regression

Coefficients ^a									
Model		Unstandardized Coefficients		Standardized Coefficients			95.0% Confidence Interval for B		Collinearity Statistics
		B	Std. Error	Beta	t	Sig.	Lower Bound	Upper Bound	Tolerance
1	(Constant)	1291933.244	8327700.888		.155	.877	-15117936.620	17701803.108	
	Duration	35611.069	51970.589	.085	.685	.494	-66797.820	138019.959	.189
	Awards Won	-59371.328	59562.757	-.116	-.997	.320	-176740.709	57998.053	.218
	Award Nominations	94508.210	39152.373	.292	2.414	.017	17357.822	171658.598	.200
	IMDb Rating	1662751.305	1086962.433	.111	1.530	.127	-479125.824	3804628.435	.558
	Type=TV Series	18579181.586	4773849.903	.489	3.892	<.001	9172232.813	27986130.360	.185
	Genre=Anime	-26160668.486	3933485.547	-.593	-6.651	<.001	-33911665.648	-18409671.325	.368
	Genre=Comedy	-11811643.155	3124671.525	-.268	-3.780	<.001	-17968859.147	-5654427.162	.583
	Genre=Drama	-14713809.858	3223493.660	-.334	-4.565	<.001	-21065756.473	-8361863.243	.548
	Genre=Horror	-9288035.564	3170849.399	-.208	-2.929	.004	-15536245.809	-3039825.319	.578
	Genre=Thriller	-8135748.379	3144130.921	-.183	-2.588	.010	-14331309.429	-1940187.330	.588
	GlobalAvailability=No	-4072393.401	1943231.741	-.121	-2.096	.037	-7901563.111	-243223.692	.873
									VIF

a. Dependent Variable: Viewership

Source: IBM SPSS Statistics 29.0

From the above table regression equation for predicting viewership is obtained as follows:

$$\text{Viewership} = 1291933.244 + 35611.069\text{Duration} - 59371.328\text{Awards Won} + 94508.21\text{Award Nominations} + 1662751.305\text{IMDb Rating} + 18579181.59 \text{ I}_1\text{Type_TV Series} - 26160668.49$$

Netflix Viewership Analysis

I₂Genre_Anime -11811643.16 I₃Genre_Comedy -14713809.86 I₄Genre_Drama -9288035.564
I₅Genre_Horror -8135748.379 I₆Genre_Thriller - 4072393.401 I₇Global Availability_No.

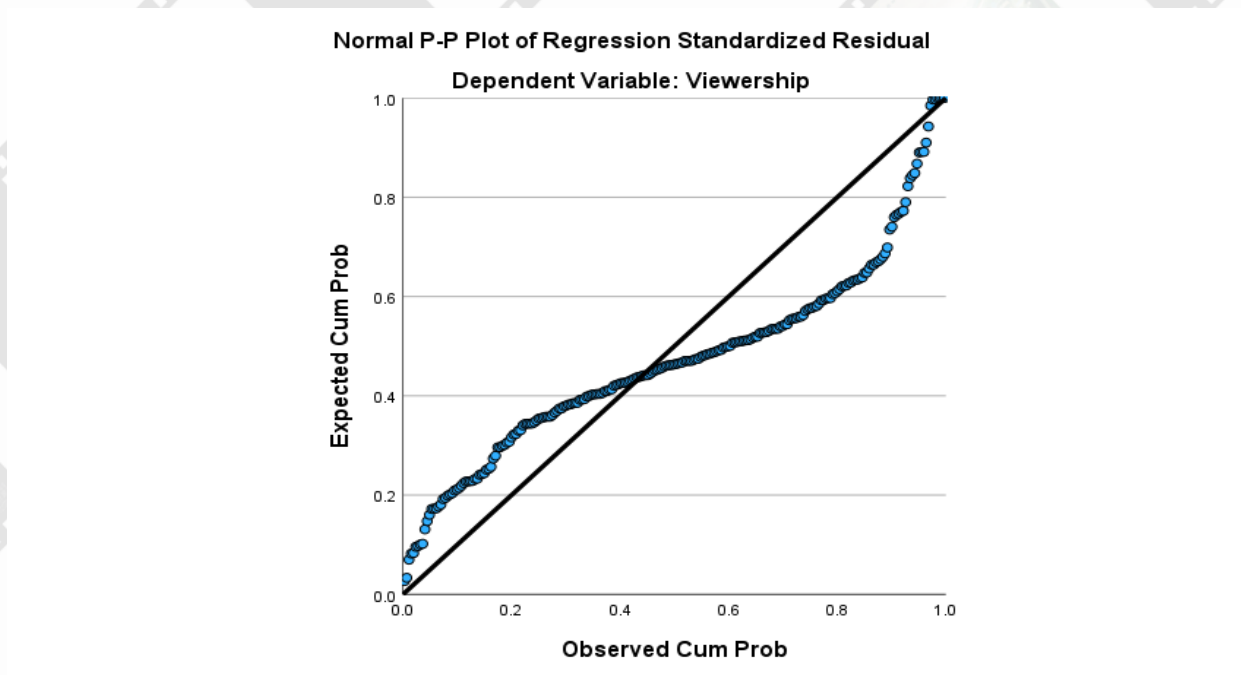
Table 5: Correlation Matrix of Multiple Linear Regression

Correlations													
	Viewership	Duration	Awards Won	Award Nominations	IMDb Rating	Type=TV Series	Genre=Anime	Genre=Comedy	Genre=Drama	Genre=Horror	Genre=Thriller	GlobalAvailability=No	
Pearson Correlation	Viewership	1.000	-.083	.174	.254	.258	.242	-.137	-.082	-.087	-.007	.064	-.189
	Duration	-.083	1.000	.221	.204	-.118	-.858	-.664	.054	.234	-.008	.219	.273
	Awards Won	.174	.221	1.000	.881	.426	-.076	-.128	-.105	.334	-.061	-.027	.058
	Award Nominations	.254	.204	.881	1.000	.489	-.033	-.148	-.086	.329	-.057	-.010	.052
	IMDb Rating	.258	-.118	.426	.489	1.000	.340	.170	-.101	.181	-.237	.084	.019
	Type=TV Series	.242	-.858	-.076	-.033	.340	1.000	.645	-.132	-.209	-.074	-.100	-.267
	Genre=Anime	-.137	-.664	-.128	-.148	.170	.645	1.000	-.202	-.202	-.199	-.199	-.062
	Genre=Comedy	-.082	.054	-.105	-.086	-.101	-.132	-.202	1.000	-.202	-.199	-.199	-.039
	Genre=Drama	-.087	.234	.334	.329	.181	-.209	-.202	-.202	1.000	-.199	-.199	.007
	Genre=Horror	-.007	-.008	-.061	-.057	-.237	-.074	-.199	-.199	-.199	1.000	-.196	-.002
	Genre=Thriller	.064	.219	-.027	-.010	.084	-.100	-.199	-.199	-.199	-.196	1.000	.090
	GlobalAvailability=No	-.189	.273	.058	.052	.019	-.267	-.062	-.039	.007	-.002	.090	1.000
Sig. (1-tailed)	Viewership	.	.100	.004	<.001	<.001	<.001	.017	.104	.090	.455	.162	.002
	Duration	.100	.	.000	.001	.034	.000	.203	.000	.450	.000	.000	.000
	Awards Won	.004	.000	.	.000	.000	.122	.024	.054	.000	.174	.337	.186
	Award Nominations	.000	.001	.000	.	.000	.308	.011	.093	.000	.191	.441	.212
	IMDb Rating	.000	.034	.000	.000	.	.000	.004	.061	.002	.000	.100	.382
	Type=TV Series	.000	.000	.122	.308	.000	.	.000	.021	.001	.128	.062	.000
	Genre=Anime	.017	.000	.024	.011	.004	.000	.	.001	.001	.001	.001	.172
	Genre=Comedy	.104	.203	.054	.093	.061	.021	.001	.	.001	.001	.001	.276
	Genre=Drama	.090	.000	.000	.000	.002	.001	.001	.001	.	.001	.001	.458
	Genre=Horror	.455	.450	.174	.191	.000	.128	.001	.001	.001	.	.001	.485
	Genre=Thriller	.162	.000	.337	.441	.100	.062	.001	.001	.001	.001	.	.083
N	GlobalAvailability=No	.002	.000	.186	.212	.382	.000	.172	.276	.458	.485	.083	.
	Viewership	238	238	238	238	238	238	238	238	238	238	238	238
	Duration	238	238	238	238	238	238	238	238	238	238	238	238
	Awards Won	238	238	238	238	238	238	238	238	238	238	238	238
	Award Nominations	238	238	238	238	238	238	238	238	238	238	238	238
	IMDb Rating	238	238	238	238	238	238	238	238	238	238	238	238
	Type=TV Series	238	238	238	238	238	238	238	238	238	238	238	238
	Genre=Anime	238	238	238	238	238	238	238	238	238	238	238	238
	Genre=Comedy	238	238	238	238	238	238	238	238	238	238	238	238
	Genre=Drama	238	238	238	238	238	238	238	238	238	238	238	238
	Genre=Horror	238	238	238	238	238	238	238	238	238	238	238	238
	Genre=Thriller	238	238	238	238	238	238	238	238	238	238	238	238
	GlobalAvailability=No	238	238	238	238	238	238	238	238	238	238	238	238

Source: IBM SPSS Statistics 29.0

The correlations among the various predictor variables can be examined from the table above to evaluate multicollinearity. While we observe multicollinearity among the independent variables, we opted not to create a separate equation to address this limitation.

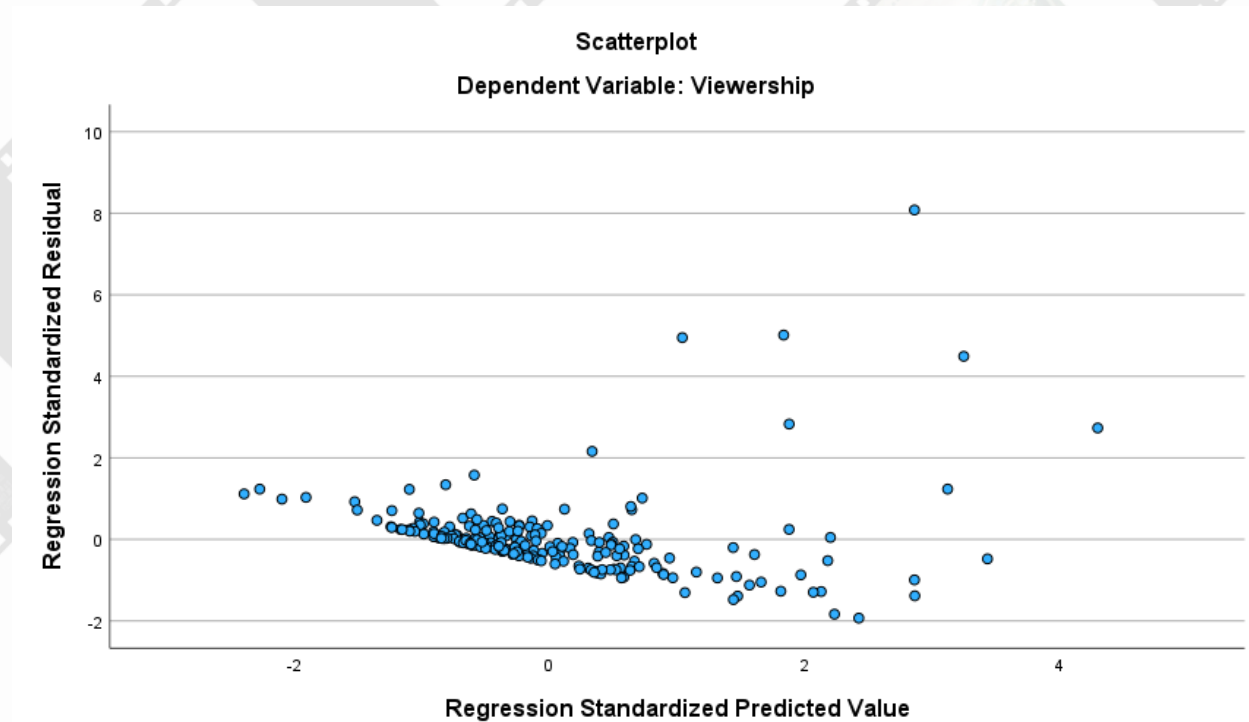
Figure 1: Normal P-P Plot



Source: IBM SPSS Statistics 29.0

Although there is evidence of non-normality in our sample, the robustness of our findings is supported by the large sample size, mitigating the impact of this limitation.

Figure 2: Scatterplot



Source: IBM SPSS Statistics 29.0

From the scatterplot, it's apparent that there's heteroscedasticity because the residuals exhibit a fan-shaped pattern. The residuals don't seem to be independent because viewers' tastes and preferences influence their choices, suggesting that there may be some underlying patterns or relationships affecting the data.

Multiple Linear Regression Conclusion

A multivariate regression was performed to investigate the variations in Viewership concerning content type and across different genres. The correlation between the variables was strong, $r(14) = 0.582$. The regression equation for predicting viewership was:

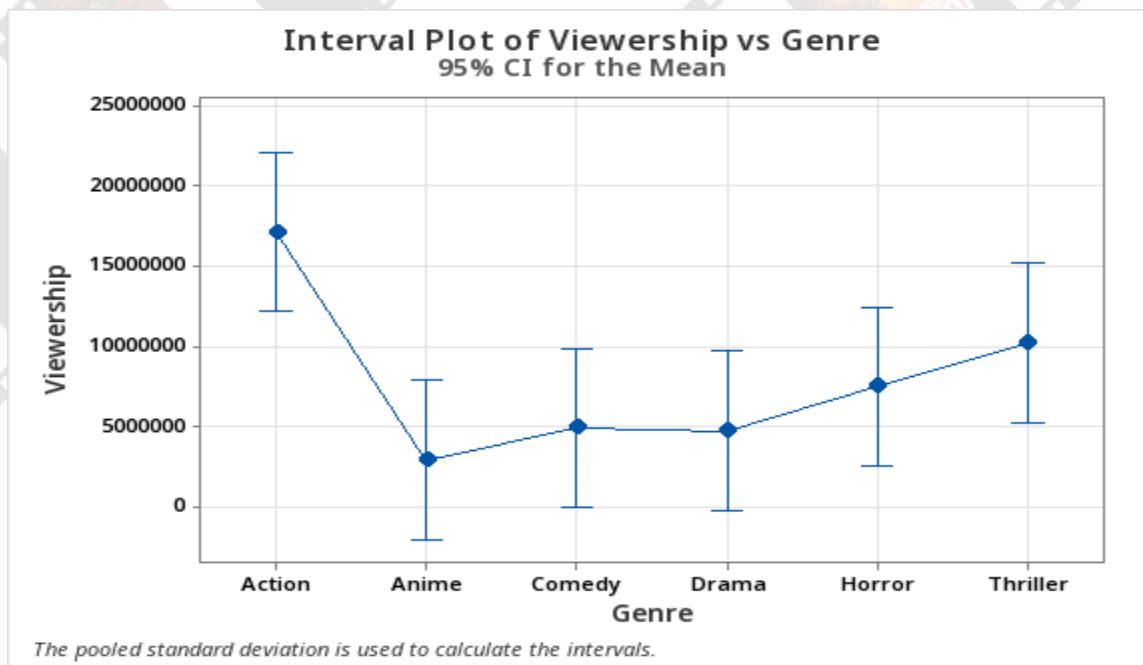
$$\begin{aligned} \text{Viewership} = & 1291933.244 + 35611.069\text{Duration} - 59371.328\text{Awards Won} + 94508.21\text{Award} \\ & \text{Nominations} + 1662751.305\text{IMDb Rating} + 18579181.59 \text{I}_1\text{Type_TV Series} - 26160668.49 \\ & \text{I}_2\text{Genre_Anime} - 11811643.16 \text{I}_3\text{Genre_Comedy} - 14713809.86 \text{I}_4\text{Genre_Drama} - 9288035.564 \\ & \text{I}_5\text{Genre_Horror} - 8135748.379 \text{I}_6\text{Genre_Thriller} - 4072393.401 \text{I}_7\text{Global Availability_No.} \end{aligned}$$

The F-statistics value of 10.528 with a P-value < 0.001 indicates that the regression equation is a very good fit for our model. The R^2 value of 0.339 implies that 33.9% of the variation in "Viewership" in our regression model can be predicted by the selected predictor variables. However, the relatively low R^2 value suggests that our model has limitations and additional predictors should be considered. The Durbin-Watson statistic of 1.014 suggests the presence of some positive autocorrelation. Although the ANOVA output indicates that the model's utility is statistically significant at the 5% level of significance (P-value < 0.001), the histogram (Figure 5) and normal PP-plot (Figure 1) of residuals shows deviations from normality. Additionally, the scatter plot (Figure 2) of residuals does not exhibit uniform distribution, indicating heteroscedasticity, and there are extreme outliers present. These factors raise concerns about the reliability of our model. Increasing the sample size may help address these issues.

Test 2: One way ANOVA

We performed a One-Way Analysis of Variance (ANOVA) test to examine whether there is a significant difference in the average viewership among different genres.

Figure 3: Interval Plot of Viewership vs Genre



Source: Minitab SSA V.01.01.2024

Netflix Viewership Analysis

The box plot suggests that the population means of viewership based on genre are not unequal as there seems to be a slight amount of overlapping.

The following are the Null and Alternative hypotheses for testing One-Way ANOVA.

H_0 : $\mu_{\text{Action}} = \mu_{\text{Anime}} = \mu_{\text{Comedy}} = \mu_{\text{Drama}} = \mu_{\text{Horror}} = \mu_{\text{Thriller}}$

H_a : At least one mean differs from the other means

Table 6: ANOVA – Viewership

ANOVA					
Viewership_Netflix					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	5368.794	5	1073.759	4.226	.001
Within Groups	59459.792	234	254.102		
Total	64828.586	239			

Source: IBM SPSS Statistics 29.0

*Viewership values measured in one million.

The ANOVA table presented above includes the Sig. value, which is essential for deciding whether to reject or fail to reject the null hypothesis.

The following are the Null and Alternative hypotheses for comparing the variances of the viewership among the genres.

H_0 : $\sigma^2_{\text{Action}} = \sigma^2_{\text{Anime}} = \sigma^2_{\text{Comedy}} = \sigma^2_{\text{Drama}} = \sigma^2_{\text{Horror}} = \sigma^2_{\text{Thriller}}$

H_a : At least one of the variances is different from the other variances.

Table 7: Levene's Homogeneity of Variances

Tests of Homogeneity of Variances					
		Levene Statistic	df1	df2	Sig.
Viewership_Netflix	Based on Mean	5.806	5	234	<.001

Source: IBM SPSS Statistics 29.0

*Viewership values measured in one million.

The table above displays the Levene's test for homogeneity of variances of Netflix viewership across different genres.

Table 8: Tukey HSD – Homogeneous Subsets

Viewership_Netflix			
Tukey HSD ^a			
		Subset for alpha = 0.05	
Genre	N	1	2
Anime	40	2.9405	
Drama	40	4.7650	
Comedy	40	4.9642	
Horror	40	7.5206	7.5206
Thriller	40	10.2283	10.2283

Netflix Viewership Analysis

Action	40		17.1529
Sig.		.320	.079

Means for groups in homogeneous subsets are displayed.

- a. Uses Harmonic Mean Sample Size = 40.000.

Source: IBM SPSS Statistics 29.0

**Viewership values measured in one million*

The Tukey HSD (Homogeneous Subsets) table indicates that the genres Anime, Drama, Comedy, Horror, and Thriller have similar mean viewership values. Likewise, Horror, Thriller, and Action exhibit comparable mean viewership values. However, the Action genre notably differs in mean viewership compared to all other genres.

Table 9: ANOVA Effect Sizes

ANOVA Effect Sizes ^{a,b}				
			95% Confidence Interval	
			Lower	Upper
		Point Estimate		
Viewership_Netflix	Eta-squared	.083	.015	.139
	Epsilon-squared	.063	-.006	.121
	Omega-squared Fixed-effect	.063	-.006	.120
	Omega-squared Random-effect	.013	-.001	.027

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

b. Negative but less biased estimates are retained, not rounded to zero.

Source: IBM SPSS Statistics 29.0

The table represents the Eta-squared Effect Size between the dependent variable – Viewership and Explanatory variable – Genre.

One-Way ANOVA Conclusion

The One-Way ANOVA test was utilized to compare the viewership of six genres. Although the box plot initially suggested that the population means of viewership based on genre are equal, the ANOVA table output revealed a p-value of 0.001, indicating a significant difference in at least one mean viewership from the others based on genre. We employed Levene's test for homogeneity to assess the equality of variances, yielding a p-value of less than 0.001, which is below the significance level of 0.05. This suggests that we reject the assumption of equal variances in viewership based on six genres.

However, the effect size, calculated using Eta Square, was 0.083, indicating that our results may not be very reliable due to a small effect size. Subsequent Tukey's test confirmed that the genres Anime, Drama, Comedy, Horror, Thriller, as well as Horror, Thriller, and Action, have similar mean viewership values. Nevertheless, the Action genre notably stands out with a different mean viewership compared to the other genres.

Test 3: Two-way ANOVA

In this test, we conducted a two-way ANOVA to evaluate whether both the type of streaming content and genre have separate effects on viewership. Additionally, we examined whether there is an interaction effect between Type and genre that influences viewership.

Below are the null and alternative hypotheses formulated to assess the effects as discussed above.

- (i) Effect of Genre on viewership

$$H_0 : \mu_{\text{Action}} = \mu_{\text{Anime}} = \mu_{\text{Comedy}} = \mu_{\text{Drama}} = \mu_{\text{Horror}} = \mu_{\text{Thriller}}$$

H_a : At least one mean differs from the other means

Since the P-value for genre is less than 0.001, we have sufficient evidence to state that Genre has an effect on viewership.

(ii) Effect of Type on viewership

$$H_0 : \mu_{\text{Movies}} = \mu_{\text{Series}}$$

H_a : At least one mean differs from the other means

Since the P-value for genre is less than 0.001, we have sufficient evidence to state that type has an effect on viewership.

(iii) Effect of Genre*Type on viewership

$$H_0 : \mu_{\text{Genre} * \text{Type}} = 0$$

$$H_a : \mu_{\text{Genre} * \text{Type}} \neq 0$$

Since the P-value for genre is less than 0.001, we have sufficient evidence to state that the interaction between genre and type has an effect on viewership.

Table 10: Tests of Between-Subjects Effects

Tests of Between-Subjects Effects						
Dependent Variable: Viewership_Netflix						
Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	24290.177 ^a	11	2208.198	12.420	<.001	.375
Intercept	19743.547	1	19743.547	111.044	<.001	.328
Type	8221.094	1	8221.094	46.238	<.001	.169
Genre	11317.864	5	2263.573	12.731	<.001	.218
Type * Genre	7109.120	5	1421.824	7.997	<.001	.149
Error	40538.410	228	177.800			

Netflix Viewership Analysis

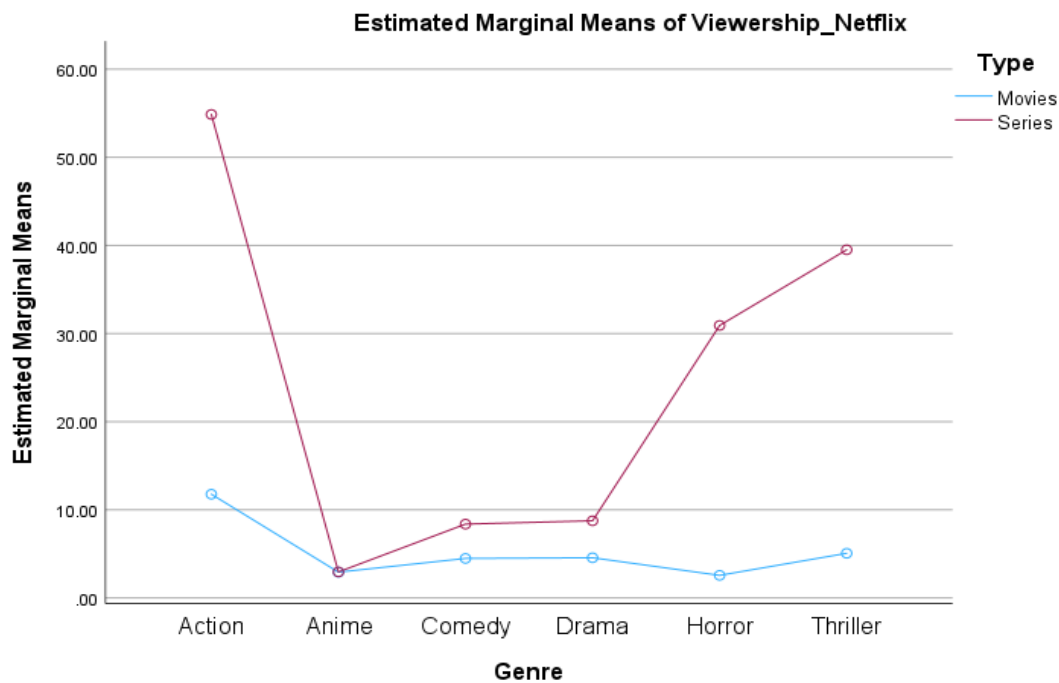
Total	79915.657	240			
Corrected Total	64828.586	239			

a. R Squared = .375 (Adjusted R Squared = .345)

Source: IBM SPSS Statistics 29.0

The above table shows P-value and the effect sizes for the relationships between viewership and type, viewership and genre, and viewership and the interaction between type and genre. The effect size values indicate the strength of these relationships.

Figure 4: Marginal Means of viewership



Source: IBM SPSS Statistics 29.0

Figure 2 represents the interaction effect of type and genre on viewership.

Two-Way ANOVA Conclusion

A Two-Way ANOVA test was conducted to compare Netflix viewership using both Type and Genre as factors. The results showed that the p-values for the simple main effects and joint effect were less than 0.001. This indicates that both Type and Genre are independently and jointly useful in explaining the variation in Netflix viewership.

Additionally, the graph associated with the ANOVA test revealed some interaction between Type and Genre. This interaction suggests that the relationship between Type and viewership can vary depending on the Genre, supporting the notion of a multiplicative model.

However, despite the statistical significance of the factors and their interaction, the effect size was found to be low at 0.37. This means that only 37% of the variation in Netflix viewership can be attributed to the variables of Type and Genre in our model. While the factors are significant, the effect size indicates that they may not fully account for all the variability in viewership, and other factors may also play a role.

Test 4: Chi-squared test for Goodness of Fit

i) Chi-squared test for Goodness of Fit - Genre

We conducted a Chi-squared test for Goodness of Fit test to determine whether there is an equal preference among the genres - Action, Anime, Comedy, Drama, Horror, and Thriller. Preference was calculated using viewership data collected from Netflix's engagement report. Additionally, we applied the same test to investigate whether there is an equal preference between Movies and Series as types of streaming content.

The following are the Null and Alternative hypotheses for testing genre preference.

H_0 : All genres - Action, Anime, Comedy, Drama, Horror, and Thriller - are equally preferred.

H_a : There is a difference in preference among the genres Action, Anime, Comedy, Drama, Horror, and Thriller.

Table 11: Goodness of Fit - Genre

	Genre		
	Observed N	Expected N	Residual
Action	687	317.8	369.2
Anime	115	317.8	-202.8
Comedy	198	317.8	-119.8
Drama	194	317.8	-123.8
Horror	304	317.8	-13.8
Thriller	409	317.8	91.2
Total	1907		

Source: IBM SPSS Statistics 29.0

**Observed and Expected viewership values measured in one million.*

The Goodness of Fit table for genre displays the anticipated and observed viewership figures, along with their corresponding residuals for each genre.

Table 12: Chi-Square Test Statistics - Genre

Test Statistics	
Genre	
Chi-Square	678.415 ^a
df	5
Asymp. Sig.	<.001

a. 0 cells (0.0%) have expected frequencies less than 5. The minimum expected cell frequency is 317.8.

Source: IBM SPSS Statistics 29.0

**Viewership values measured in one million.*

The above table displays the Chi-Square statistic, degrees of freedom, and P-value for the weighted viewership categorized by Genre.

Conclusion I

We used a Chi-squared test to check if Netflix users' preferences for movie and series genres were random. In each category, the expectation to see at least 5 instances was satisfied. The gap between our expectations and the actual observations was substantial. This significant difference

led us to a crucial conclusion: there is indeed a variation in preference among the genres Action, Anime, Comedy, Drama, Horror, and Thriller. The Chi-squared statistic (χ^2) of 678.415, with a p-value of less than 0.001, signifies that Netflix users strongly favored Action genre movies and series over the other genres.

ii) Chi-squared test for Goodness of Fit – Type

The following are the Null and Alternative hypotheses for testing genre preference.

H₀: There is no significant difference in preference between Movies and Series.

H_a: There is a significant difference in preference between Movies and Series.

Table 13: Goodness of Fit – Type

	Type		
	Observed N	Expected N	Residual
Movies	1016	953.5	62.5
Series	891	953.5	-62.5
Total	1907		

Source: IBM SPSS Statistics 29.0

**Observed and Expected viewership values measured in one million.*

The Goodness of Fit table for type represents the expected and observed viewership along with the corresponding residual for movies and series.

Table 14: Chi-Square Test Statistics - Type

Test Statistics	
Type	
Chi-Square	8.193 ^a
df	1
Asymp. Sig.	.004

a. 0 cells (0.0%) have expected frequencies less than 5. The minimum expected cell frequency is 953.5.

Source: IBM SPSS Statistics 29.0

**Viewership values measured in one million.*

The above table displays the Chi-Square statistic, degrees of freedom, and P-value for the weighted viewership categorized by Type.

Conclusion II

A Chi-squared goodness of fit test was utilized to determine if Netflix users' preferences for streaming content type - movies or series - differed significantly from randomness. Expected frequencies in all cells were greater than 5, meeting the statistical criterion. The substantial difference between expected and observed values led to a significant conclusion that there is a significant difference in preference between Movies and Series. The χ^2 statistic (df = 5, N =

1907) equalled 8.193, with a corresponding p-value of 0.004. This indicates that Netflix viewers preferred movies, likely because they can be watched entirely in a short period of time.

Our Conclusion and Recommendations

A multivariate regression was conducted to understand the variations in Netflix viewership based on content type and genres. The correlation between variables was strong, indicating a relationship between them. The regression equation showed that viewership could be predicted using factors like Duration, Awards Won, Award Nominations, IMDb Rating, Type (TV Series), and various Genres.

The F-statistics value of 10.528 with a p-value < 0.001 indicated that the regression model was a good fit for predicting viewership. However, the relatively low R-squared value of 0.339 suggested that only 33.9% of the variation in viewership could be explained by the selected predictors. This indicates that other factors not included in the model might also influence viewership.

Additionally, there were concerns about the model's reliability due to positive autocorrelation, deviations from normality in residuals, heteroscedasticity, and the presence of extreme outliers. Increasing the sample size may help address these issues.

An ANOVA test comparing viewership among different genres showed a significant difference in mean viewership based on genre. However, the effect size was relatively small, suggesting limited reliability in the results.

A Two-Way ANOVA test revealed a strong joint effect of Type and Genre on Netflix viewership, indicating that these factors independently and jointly influence viewership.

A Chi-squared test showed that Netflix users' preferences for movie and series genres were not equal, with a significant preference for the Action genre. Another Chi-squared test indicated a significant difference in preference between Movies and Series, with viewers showing a preference for movies.

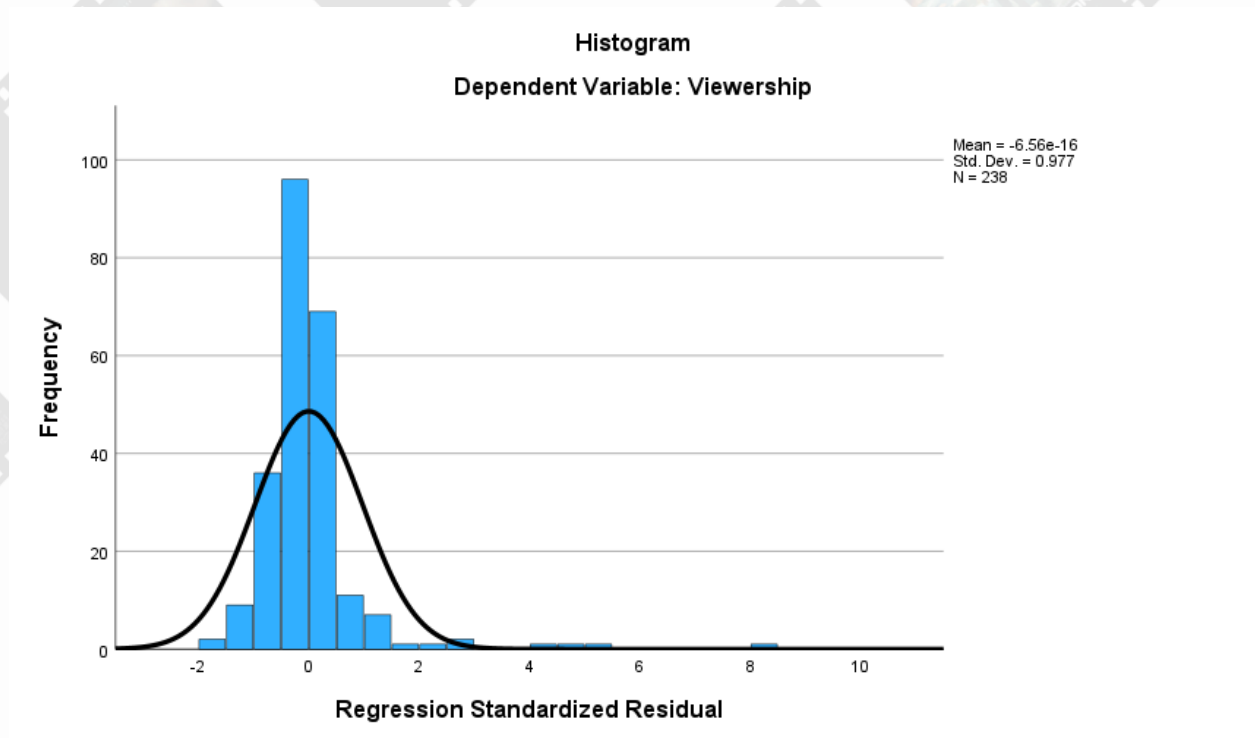
In summary, the analysis identified significant relationships between viewership and factors like content type, genres, and specific variables. However, the models have limitations, and additional factors such original language, audio and subtitle languages available, content ratings, critic reviews and time of release should be considered for a more comprehensive understanding of Netflix viewership patterns.

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Netflix Viewership Analysis

Appendix A: Figure 5 - Histogram depicting Normality



Source: IBM SPSS Statistics 29.0

Netflix Viewership Analysis

Appendix B: Table 15 - Multiple Comparisons – ANOVA

Multiple Comparisons

Dependent Variable: Viewership_Netflix

Tukey HSD

(I) Genre	(J) Genre	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Action	Anime	14.2124*	2.98161	<.001	5.6429	22.7819
	Comedy	12.1887*	2.98161	<.001	3.6192	20.7582
	Drama	12.3879*	2.98161	<.001	3.8184	20.9574
	Horror	9.6323*	2.98161	.018	1.0628	18.2018
	Thriller	6.9246	2.98161	.189	-1.6449	15.4941
Anime	Action	-14.2124*	2.98161	<.001	-22.7819	-5.6429
	Comedy	-2.0237	2.98161	.984	-10.5932	6.5458
	Drama	-1.8245	2.98161	.990	-10.3939	6.7450
	Horror	-4.5801	2.98161	.641	-13.1496	3.9894
	Thriller	-7.2878	2.98161	.146	-15.8573	1.2817
Comedy	Action	-12.1887*	2.98161	<.001	-20.7582	-3.6192
	Anime	2.0237	2.98161	.984	-6.5458	10.5932
	Drama	.1992	2.98161	1.000	-8.3703	8.7687
	Horror	-2.5564	2.98161	.956	-11.1259	6.0131
	Thriller	-5.2641	2.98161	.490	-13.8336	3.3054
Drama	Action	-12.3879*	2.98161	<.001	-20.9574	-3.8184
	Anime	1.8245	2.98161	.990	-6.7450	10.3939
	Comedy	-.1992	2.98161	1.000	-8.7687	8.3703
	Horror	-2.7556	2.98161	.940	-11.3251	5.8139
	Thriller	-5.4633	2.98161	.447	-14.0328	3.1062
Horror	Action	-9.6323*	2.98161	.018	-18.2018	-1.0628
	Anime	4.5801	2.98161	.641	-3.9894	13.1496
	Comedy	2.5564	2.98161	.956	-6.0131	11.1259
	Drama	2.7556	2.98161	.940	-5.8139	11.3251
	Thriller	-2.7077	2.98161	.944	-11.2772	5.8618
Thriller	Action	-6.9246	2.98161	.189	-15.4941	1.6449
	Anime	7.2878	2.98161	.146	-1.2817	15.8573
	Comedy	5.2641	2.98161	.490	-3.3054	13.8336
	Drama	5.4633	2.98161	.447	-3.1062	14.0328
	Horror	2.7077	2.98161	.944	-5.8618	11.2772

Based on observed means.

The error term is Mean Square(Error) = 177.800.

*. The mean difference is significant at the .05 level.

Source: IBM SPSS Statistics 29.0

*Viewership values measured in one million.

Appendix C: References

- Davis, B. (2020). *Business Analytics: Data Analysis and Storytelling for Business*. Idaho: Edify.
- Duarte, F. (2024). *Exploding Topics*. Retrieved from Exploding Topics: <https://explodingtopics.com/blog/video-streaming-stats>
- IMDb. (2023). *IMDb*. Retrieved from IMDb: <https://www.imdb.com/>
- Netflix. (2023). *What We Watched: A Netflix Engagement Report*. Retrieved from Netflix: <https://about.netflix.com/en/news/what-we-watched-a-netflix-engagement-report>
- Netflix. (2024). *Netflix*. Retrieved from Netflix: <https://www.netflix.com>
- Wikipedia. (2023). *Wikipedia*. Retrieved from Wikipedia: <https://www.wikipedia.org/>